

Nav Theory

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December 16, 2025

Geographic Position Rate from Velocity and Heading

Let the position be (φ, λ) , where

- φ = latitude (rad),
- λ = longitude (rad).

Let v be the horizontal speed (m/s) and ψ the heading/COG (rad), measured clockwise from north (N=0, E= $\pi/2$).

1) Velocity Components in Local ENU Frame

$$v_N = v \cos \psi, \quad v_E = v \sin \psi$$

2) Conversion to Angular Rates (WGS-84)

$$\dot{\varphi} = \frac{v_N}{M(\varphi)} = \frac{v \cos \psi}{M(\varphi)}$$
$$\dot{\lambda} = \frac{v_E}{N(\varphi) \cos \varphi} = \frac{v \sin \psi}{N(\varphi) \cos \varphi}$$

where (WGS-84 ellipsoid):

$$e^2 = f(2 - f), \quad f = \frac{1}{298.257223563}, \quad a = 6378137 \text{ m}$$

$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}$$

$$M(\varphi) = \frac{a(1 - e^2)}{(1 - e^2 \sin^2 \varphi)^{3/2}}$$

3) Output in Vector Form

$$\begin{bmatrix} \dot{\lambda} \\ \dot{\varphi} \end{bmatrix} = \begin{bmatrix} \frac{v \sin \psi}{N(\varphi) \cos \varphi} \\ \frac{v \cos \psi}{M(\varphi)} \end{bmatrix} \quad (\text{rad/s})$$

For degrees per second:

$$\dot{\lambda}_{\text{deg}} = \dot{\lambda} \cdot \frac{180}{\pi}, \quad \dot{\varphi}_{\text{deg}} = \dot{\varphi} \cdot \frac{180}{\pi}$$

Note: These formulas are valid locally (small steps) and assume heading is the true course over ground (COG true).